What is a Dependency?

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Definition

A dependency is a relationship between attributes where the value of one attribute determines the value of another attribute. Think of an [Aadhaar Redacted] number. If someone tells you an [Aadhaar Redacted] number, you can identify exactly one person's details.

Without Dependencies

You cannot:

- Find Candidate Keys.
- Normalize tables.
- Detect Partial Dependencies.
- Detect Transitive Dependencies.
- Verify BCNF.

Dependencies are the language of normalization.

Determinant

Definition: A Determinant is the attribute (or set of attributes) that determines another attribute.

Dependency: $RollNo \rightarrow Student$
Determinant: **RollNo**

Composite Determinant

Sometimes multiple attributes together determine another.

Example: $(StudentID, CourseID) \rightarrow Grade$

Dependent Attribute

Definition: A Dependent Attribute is the attribute whose value is determined by another attribute. here grade.

Types of Dependencies

Normalization uses several kinds of dependencies. We'll study each briefly here and then dedicate full chapters to them later.

1. Functional Dependency (FD)

Definition: One attribute uniquely determines another.

Notation: $X \rightarrow Y$

Meaning: If two rows have the same X, they must also have the same Y.

Example: $StudentID \rightarrow Name$

Full Functional Dependency

An attribute depends on the entire composite key.

Example: $(StudentID, CourseID) \rightarrow Grade$

Partial Dependency

An attribute depends on only part of a composite key.

Example:

- Composite Key: $(StudentID, CourseID)$

- Dependencies: $StudentID \rightarrow StudentName$

StudentName depends only on StudentID. It does not require CourseID. This is a Partial Dependency.

Transitive Dependency

One non-key attribute determines another non-key attribute.

Example:

- $EmployeeID \rightarrow DepartmentID$
- $DepartmentID \rightarrow Manager$
- Therefore: $EmployeeID \rightarrow Manager$

Partial dependencies violate Second Normal Form (2NF). Transitive dependencies violate Third Normal Form (3NF).

Multivalued Dependency (MVD)

One attribute independently determines multiple values of another attribute.

Notation: $X \twoheadrightarrow Y$

Join Dependency (JD)

A table can be decomposed into multiple tables and reconstructed using joins without losing information. This forms the basis of Fifth Normal Form (5NF).

Dependency Hierarchy

```
Dependency
|
├── Functional Dependency (FD)
|      |
|      ├── Full Functional Dependency
|      ├── Partial Dependency
|      └── Transitive Dependency
|
```

```
├── Multivalued Dependency (MVD)
│
└── Join Dependency (JD)
```

A relationship connects entities. A dependency connects attributes.